A Design of Smart Charging Architecture for Battery Electric Vehicles

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2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE) | 979-8-3503-9678-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ISGTEUROPE56780.2023.10408308

Abstract*— The effect of operating many electric vehicle (EVs) charging stations (CSs) should not negatively impact the electric power distribution system, primarily when many CSs are used simultaneously in one location. A smart charging controller can regulate the charging pattern to fulfill EV owners' needs while considering the electric power availability and readiness of the feeder. Many scientific publications on smart charging have discussed charging strategies; however, studies on smart-charging architectures are scarce. This paper proposes a new design for a smart charging architecture for battery EVs by integrating an existing charging station management system (CSMS) with an electrical power monitoring system (EPMS). We constructed an EPMS using the Internet of Things (IoT), an abundantly available and inexpensive device. Experimental results show that the CSMS can communicate with CSs and the electric feeder through the open charge point protocol (OCPP) and message queuing telemetry transport (MQTT) protocol. Using the proposed smart-charging controller, the electric power system, EVs, and EV owner datasets can optimize the charging profile to maintain the reliability of the electrical grid.

Keywords: charging station, electric vehicle, IoT, smart charging

I. INTRODUCTION

Charging stations (CSs) are essential in promoting electric vehicle (EV) adoption. The CS communicates with the EV through the proprietary protocol and Charging Station Management System (CSMS) through the open charge point protocol (OCPP). The information communication technology (ICT) infrastructure for e-mobility includes networks, data communication systems, mobile applications, and back-end systems: OCPP servers, central control systems or CSMS, database systems, billing systems, and payment gateways.

With an increasing number of CSs, the possibility of affecting the electrical grid stability during peak loads must be examined. The Internal Renewable Energy Agency (IRENA) [1] and the European Network of Transmission System Operators for Electricity (ENTSO-E) encourage the development of EVs, leading to the integration of EVs and electricity systems [2]. Smart charging has become a hot topic in anticipation of increasing EV use. Several researchers have published studies on smart charging. Mangipinto A. et al. simulated the charging pattern and EVs mobility [3]. They developed a model using empirical data from open-vehicle usage combined with social, economic, and weather data.

Most of the previously published articles regarding smart charging for EVs have focused on charging strategies [4], [5]. However, they did not describe the architecture and devices in detail. An exciting article disclosed an open-source package for smart charging [6] using an algorithm based on pre-computed schedules [7]. However, they did not monitor current, voltage, or power from the electrical grid. An adaptive charging network was proposed and implemented in a more complex smart charging garage [8]. It comprises 78 pieces of EV supply equipment (EVSE). They used a ZigBee-based mesh network to communicate among the CSMS, sensors, and EVSEs. The sensors measured the power, current, and voltage within the local electrical network to monitor the system state and track the energy usage. They also provided feedback regarding the control algorithm.

An Internet of Things (IoT)-based centralized control strategy was proposed to coordinate EVs and distributed energy sources (DES) [9]. The control approach requires a controller at each node or measurement point with a measuring meter and a switch box to sequence phases. Each switch consists of a ZigBee wireless receiver to receive control information, snubber circuit, TRIAC, and over-voltage protection. The measuring meter consisted of various sensors installed at every node, feeder, and connection point at the substation. The measurement result is sent to the central controller using the ZigBee transceiver through message queuing and elementary transport communication protocol.

This article proposes a new architecture for the smart charging of EVs using the message queuing telemetry transport (MQTT) protocol. The aim is to develop a smart charging controller and its data communication system to increase the

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reliability of the CS infrastructure. This architecture enables the CSMS to be optimized from two perspectives: the grid operator and consumer. It provides cost benefits because it uses IoT devices that are abundantly available in the market. This article also reports preliminary implementation results for an actual CS.

II. Smart Charging

A. Definition and Benefit of Smart Charging

Referring to IRENA (2019), smart charging is defined as adapting the charging cycle of EVs to both the conditions of the electrical power system and the needs of the EV users [1]. The smart charging process can be implemented by monitoring, managing, and limiting the use of the CS to optimize energy consumption and charging time selection based on favorable tariffs. The power consumption profile is controlled using several methods, including shifting other loads or CS start settings. Hence, the electric load profile becomes flatter [10].

Based on power management, smart charging can be divided into two systems, namely, user-managed charging (UMC) and supplier-managed charging (SMC) [11]. For the UMC, it is generally more about selecting the electricity rates and charging times. The user fills the electricity and energy requirements. Most likely, users charge EVs when the rates are low, such as during off-peak load times. However, simultaneous use of multiple EVs charging can cause a sudden increase in the electrical load. At the SMC, charge decisions are made based on several conditions, such as the available real-time energy production, local energy consumption, and charge status information from nearby EVs and other electrical devices. Ideally, the CSMS can adjust the EV charging peak to match the availability of electric power and the electricity demand during peak load times [12].

B. OCPP

The OCPP was developed by the Open Charge Alliance (OCA). It is a de facto standard implemented in various CS products on the market. OCPP stipulates that a Charge Point (the technical term for a CS) communicates with the Central System when charging EVs [13].

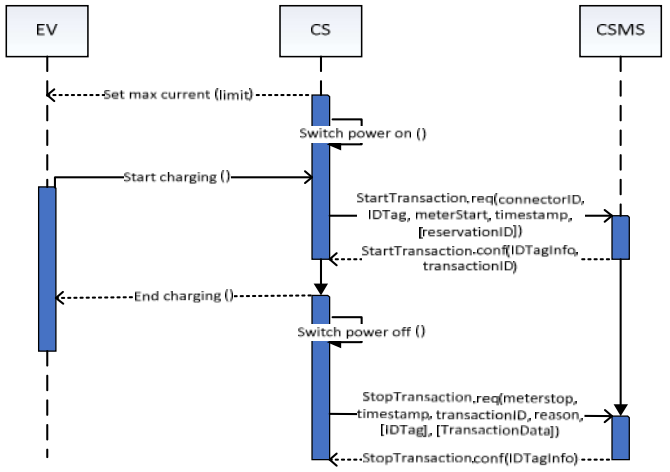


Fig. 1. The EV Charging Process Sequence [13]

The OCPP defines a set of messages categorized as a Core Profile (mandatory), Local Auth List Management Profile (optional), Reservation Profile (optional), and Smart Charging Profile (optional). The interaction between the Charge Point and the Central System begins with a connection request from the Charge Point to the Central System. It then sends a BootNotification() containing configuration data (e.g., vendor, model, type, serial number, etc.) from the CS. BootNotification() is one of the Core Profile elements and a mandatory profile in the OCPP. The Central System must respond to BootNotification() to indicate whether a Charge Point can be accepted or rejected. After the initialization process is complete, the CS is ready to be used. Fig. 1 illustrates the EV charging process sequence according to the OCPP.

C. OSCP

OCA officially released Open Smart Charging Protocol (OSCP) 2.0 in October 2020. This protocol is an enhancement of OSCP 1.0 to use flexible energy sources based on available capacity, primarily intended for smart charging EVs. It was developed in Europe, where the electricity market is based on a market mechanism with many parties involved in electricity transmission, distribution, and retail. The information flow and parties involved in the smart charging scheme with the OSCP protocol are shown in Fig. 2 [14].

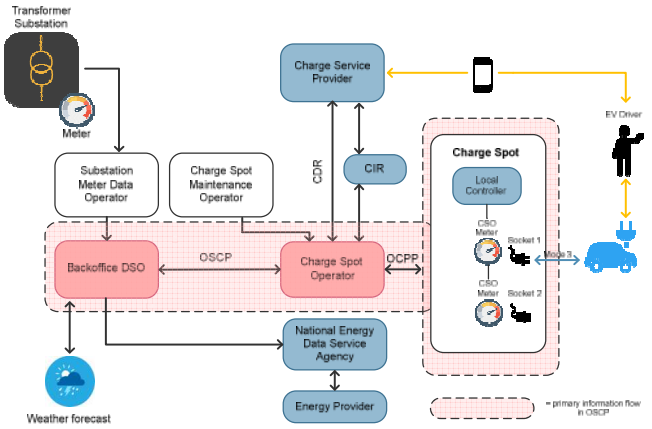


Fig. 2. Flow information and parties involved in OSCP [14]

One of the main aspects of smart charging is the prediction of future power requirements based on available resources. The forecasting mechanism requires power and the provision of power for charging, and the actors involved are described in the document specifications of OSCP 2.0. It is important to note that an EV smart controller that integrates CSMS data with the grid and server based on OSCP is not yet available on the market.

D. MQTT

MQTT is a standard-based messaging protocol or set of rules used for machine-to-machine communication. Smart sensors, wearables, and other IoT devices typically send and receive data over networks with limited resources and bandwidths. MQTT is used for data transmission because it is easy to implement and is efficient. MQTT supports messaging between the devices to the cloud and the cloud to the devices.

MQTT is used in various industries such as automotive, manufacturing, telecommunications, oil, and gas. Its advantages include being lightweight and efficient, two-way communication, scaling to millions of things, reliable messaging, support for unreliable networks, and security enabled. This study attempts to implement a smart-charging design architecture using MQTT for these reasons.

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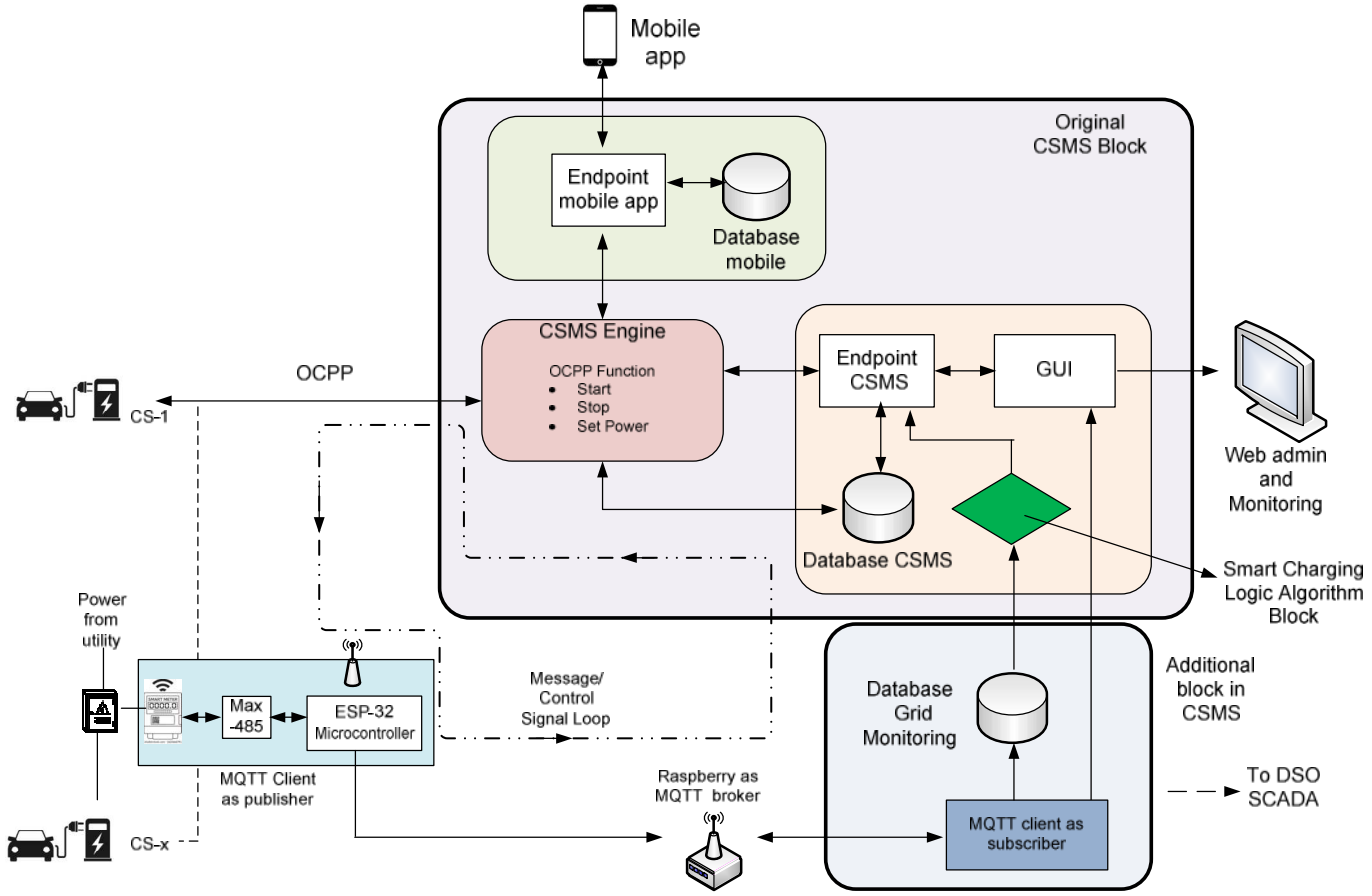


Fig. 3. Configuration of the proposed smart charging system

III. Proposed Smart Charging Architecture

Previously, we developed a CSMS by integrating five CSs from several locations [15]. In the existing CS control system, the CSMS can only control the CS without being able to monitor the feeder condition. In contrast, this paper proposes a smart CS control system in which a smart charging controller integrates an electrical power monitoring system (EPMS) and an existing CSMS, as illustrated in Fig. 4. Thus, the charging process can be performed optimally while maintaining the stability of the power grid.

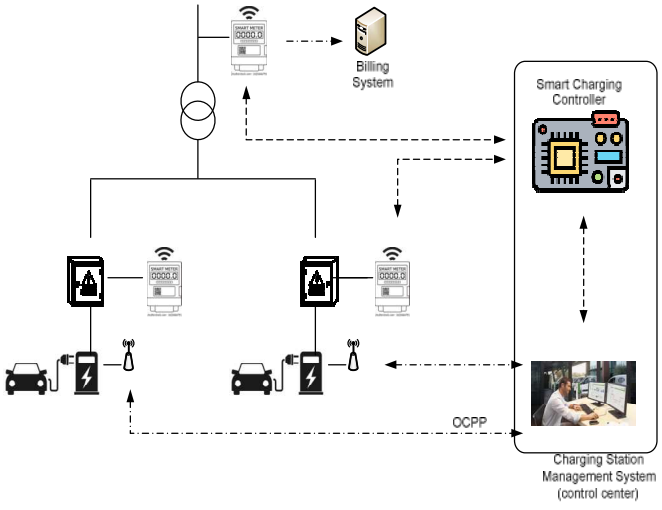


Fig. 4. Integrated EPMS and CSMS.

The installation of a CS in an electric power distribution system, as observed from the closest connection point, can generally be divided into two configurations:

a) Installation using an existing nondedicated transformer. The CS uses the available power margin from an existing transformer. Usually, this installation is implemented in buildings or when only one to two charging units are required.

b) Installation of dedicated transformer to serve CSs. Usually, this installation serves large numbers of charging units.

The maximum power that a CS can use from both A and B configurations is limited by the capacity of the transformers, cables, and circuit breakers.

Fig. 3 shows the configuration of the proposed system. The EPMS consists of digital power meters, an RS-485 converter, a microcontroller ESP-32 for power metering, and a Raspberry microcontroller as the main controller. Electric data flows from the digital power meter to the CSMS over MQTT, whereas the message flows from CSMS to CS over OCPP. The Raspberry Pi and ESP32 microcontrollers were used as the MQTT broker and client (publisher), respectively. The CSMS plays the role of MQTT client (subscriber). ESP32 reads electric data over RS485 using the MAX485 module and connects to Raspberry Pi over WiFi.

Fig. 5 shows the process flow for publishing electric data to MQTT. When starting up, the MQTT publisher connects to the WiFi and MQTT broker. Then, it reads the electric data by

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sending requests to the MODBUS address for the time, voltage, current, power, frequency, and power factor. After obtaining the complete data, it arranges the dataset in JavaScript Object Notation (JSON) format. Next, it publishes the JSON data to the MQTT broker with the topic, for example, "energyMeterJson." The MQTT publisher loops this process continuously at one-second intervals. The CSMS receives electric data from the MQTT and then processes and stores it in the database. The CSMS executes a smart-charging algorithm based on electrical data.

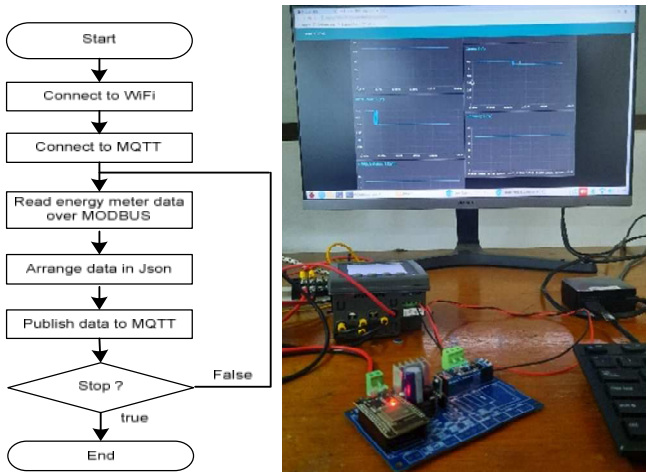


Fig. 5. Process flow of publishing data to MQTT

From the power measured at the feeder, P_f, and each EV charging plug, P_i, we can calculate the power available for charging the ego EV, P_e.

$$P_e = P_f - \sum_{i=1}^{N-1} P_i \quad (1)$$

Where N denotes the total number of the EVs being charged. The main controller also calculates the feeder's available power, as follows :

$$P_f = I_c V \quad (2)$$

Where I_c is the cable's maximum current carrying capability. Comparing the maximum cable current conductibility and the actual measurement current, I_c, we can get the % loading, L_c.

$$L_c = \frac{I_m}{I_c} \times 100\% \quad (3)$$

In order to make all charger operate without curtailment, the total charging power at the CS must be less than the available power.

$$\sum_{i=1}^{N} P_i < P_f \quad (4)$$

Fig. 6 shows the conceptual design of a smart charging algorithm that uses a dataset from charging reservations, EVs, available power from the grid, and charging profile database. This block was built using a modified CSMS (green box in Fig. 4). Charging profile data were obtained from the measurement results and modeled mathematically. Several conditions may occur based on data from the current and future use of CS and their relationship to power availability, as shown in Table 1.

Several operations and messages can be used to implement smart charging by using OCPP. Some of the messages that the CSMS can use include the SetChargingProfile and GetCompositeSchedule.

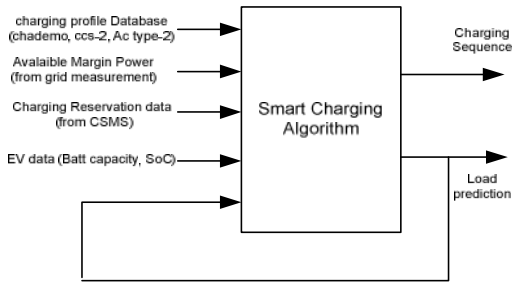


Fig. 6. Concept of smart charging algorithm

TABLE 1. Decission Table

Current condition	Future condition	Action
Low load	Low demand	Standard operation
Low load	High demand	Prepare load management
High load	Low demand	Load management
High load	High demand	Load management with load reduction

Several messages interoperate with the data within these two messages, such as charging duration, connectors, status, schedule, electrical parameters, and start and stop transactions. In addition, in the CS, there are arrangements regarding smart charging in the form of scheduling, tariffs, and internal control based on power or current. The CSMS gives commands to perform smart charging to the CSs, and the internal control system manages according to the CSMS orders. The scheme for sending and managing smart charging involving CSs and CSMS is shown in Fig. 7.

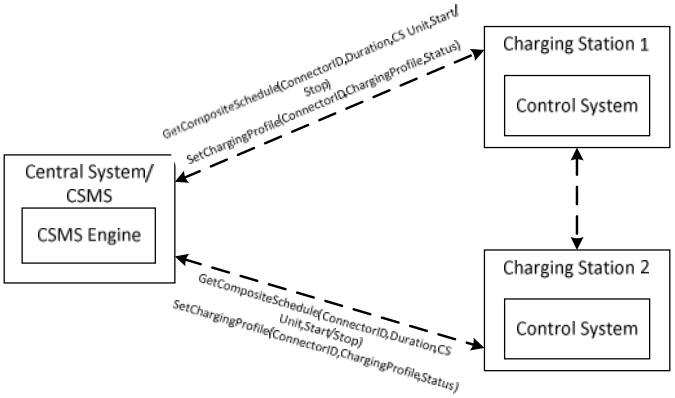


Fig. 7. Message interaction between CS and CSMS in the operation of smart charging

IV. RESULT

A prototype EPMS was developed based on the proposed architecture, as shown in Fig.3. The main components are digital energy meters, Max-485, and ESP32, which are used for reading electrical data from the grid. A data feeder reading test via an energy meter and the data acquisition system shown in Fig. 5 were successfully carried out. The electrical system data are read from the energy meter, processed on the ESP-32, and transmitted via Wi-Fi (or LAN) to the Raspberry as the main controller. The read data were then displayed on an HMI screen for monitoring. This data can also be read using an Android device. An additional block function in the CSMS that receives data transfers from the Raspberry was also successfully tested. These data are then processed by the smart-charging algorithm to give commands to the CSs.

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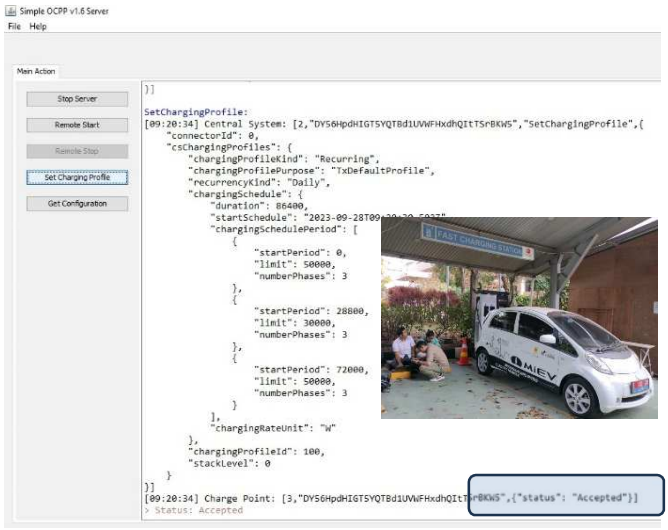


Fig. 9. Data and message transfer between the CSs and the CSMS

Sending and reading data on CSs and CSMS are tested using scheduling messages. In this message, there is a command message to schedule based on the duration, current or power rating, connectorID, and status.

To ensure that the CS can respond to commands to change the charging power, we performed testing using a CSMS tester, as shown in Fig. 9. This testing utilizes a unit data protocol called SetChargingProfile provided by the OCPP to limit the DC charging power to 30 kW between 08:00h and 20:00h, down from a daily basis of 50 kW. The test results show that the CS responded by providing an "Accepted" signal to the command/message sent by the CSMS.

From these messages, we can ensure that the CSMS is working to provide a signal to the CS based on the input from the EPMS data-processing results. The result of this experiment validates the data processing of the control-loop signal shown in Fig. 3 works properly.

V. CONCLUSION

The architectural design and prototype of a smart charging system for EVs have been described, combining the EPMS based on MQTT with the CSMS back-end system and modifications to the OCPP protocol (OCPP Extension). Electrical data from the grid were successfully gathered and processed through the developed prototype, then transmitted to the main controller via WiFi. This data was subsequently received and processed in the smart charging functional block of the CSMS. Several communication features between the EPMS and the CSMS were successfully tested.

Future work includes installing the prototype at the main distribution panel of a selected CS, recording the time series dataset of the grid electrical variables and EVs' charging profiles, developing smart charging algorithms, and implementing them at the CS for performance evaluation.

ACKNOWLEDGEMENT

The authors would like to thank to the National Research and Innovation Agency (BRIN) and PT Transisi Enjiniring Indonesia for their support to facilitate author to the ISGT Conference and overall personnel for their assistance and support throughout all steps of our research and for their help in writing the manuscript as well. The authors also acknowledge the Head of the Research Center for Energy Conversion and Conservation (PRKKE-BRIN) who supported this study.

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